

Weekly Research Report — Week 8

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Summary of Work Done

The hybrid architecture was extended to medical visual question answering, which produced the project’s first clearly significant result. Full statistical analysis was completed across both benchmarks, and a decoding-configuration confound was identified and corrected. The paper draft was completed.

Medical visual question answering

The MedQA ceiling effect suggested the hypothesis should be tested where parametric knowledge is genuinely insufficient. Medical imaging is such a setting: answering a pathology question requires reading the image, which no amount of text pretraining supplies.

Two datasets were evaluated. **VQA-RAD** (radiology, 251 yes/no test questions, Claude Opus 5) gave baseline 60.2%, GraphRAG 63.7%, hybrid 78.5%. **PathVQA** (pathology, 1,010 yes/no questions, Claude Haiku) was adopted for the paper as the larger and more defensible sample.

A task-specific knowledge graph was built from the 19,654 PathVQA training examples by rule-based extraction, yielding 6,614 entities and 19,613 relations linking visual concepts — anatomy, modality, findings — to observations.

Results

Architecture	PathVQA	vs baseline
Baseline, vision only	68.0%	—
Flat-vector RAG	70.4%	p = 0.21, n.s.
Static GraphRAG	69.9%	p = 0.26, n.s.
Hybrid GraphRAG	79.6%	p < 0.001

An 11.6 percentage point gain, highly significant: 216 questions answered correctly by the hybrid system and incorrectly by the baseline, against 99 in the reverse direction.

The components are individually inert and jointly effective. Neither flat-vector RAG nor static GraphRAG improves significantly on the baseline alone. Only the combination clears significance. The retrieval strategies recover different questions, and only together do they cover enough ground to matter. This is a stronger claim than a graded improvement across methods, and it is the paper’s central contribution.

An attempt to improve on this by combining the task-specific VQA graph with PrimeKG (43,922 entities) performed worse, not better. Domain alignment between graph and task matters more than graph scale.

Statistical analysis

McNemar’s paired test and 95% Wilson confidence intervals computed for every comparison on both benchmarks.

Architecture	MedQA (n=1,273)	PathVQA (n=1,010)
Baseline	96.0%	68.0%
Flat-vector RAG	96.5% (p=0.34)	70.4% (p=0.21)
Static GraphRAG	95.8% (p=0.72)	69.9% (p=0.26)
Hybrid GraphRAG	96.2% (p=0.72)	79.6% (p<0.001)

Decoding-configuration confound

The MedQA baseline’s per-question output file was lost in a session reset, so the run was repeated to enable significance testing. It returned **94.7%**, not the 96.3% previously recorded — and against that baseline all three retrieval systems appeared *significantly better* (p = 0.0003, 0.035, 0.0042).

That would have reversed the paper’s MedQA conclusion. It was wrong.

The re-run used a 10-token output limit with reasoning disabled, while the three retrieval architectures had run with a 2,000-token limit and the default reasoning configuration. The comparison was measuring decoding configuration,

not retrieval. The tell was that the effect was near-constant across three architecturally distinct systems — a genuine retrieval effect would vary with retrieval quality.

Re-running the baseline under settings identical to the other architectures restored it to **96.0%** and every comparison returned to non-significant ($p = 0.34, 0.72, 0.72$).

All reported comparisons now use architectures configured identically within each benchmark, and this is documented in the Evaluation Protocol section of the paper. Had the mismatched comparison been published, a reviewer asking whether settings were matched would have unravelled the result.

Paper

Complete through Abstract, Introduction, Related Work, Methodology, Results, Discussion and Conclusion. Every accuracy, confidence interval and p-value is computed from saved per-question outputs rather than reported from memory. Figures regenerated to match.

Position

The paper now has a coherent argument. Retrieval augmentation earns its cost precisely where the backbone model's parametric knowledge is insufficient, and not otherwise. The MedQA ceiling effect and the PathVQA gain are two halves of the same finding rather than a failure followed by a success.

Plan for next period

1. Instrument retrieval-stage metrics — context precision and recall, faithfulness to retrieved evidence, hallucination rate — addressing the trustworthiness direction raised by supervisors
2. Verify remaining references against source PDFs
3. Publish the cleaned notebook, result files and knowledge graphs to the public repository